Section-Aware Multi-Vector Retrieval and Hybrid Ranking for Bidirectional CV–Job Matching in Freelance Platforms

Anonymous Authors[1]

Department of Computer Science and Engineering
research@example.edu

Abstract. Matching freelance candidate profiles (Curriculum Vitae, CV) with Job Descriptions (JD) is a critical operational bottleneck in online freelance marketplaces. Traditional automated recruitment systems predominantly rely on either lexical keyword matching or single-vector dense retrieval. However, single-vector representations aggregate the entire document into an unweighted embedding, diluting fine-grained signals across distinct sections such as technical skills, practical experience, and education. Moreover, real-world recruitment involves hard and soft business constraints (e.g., mandatory skill overlap, budget compatibility, and experience thresholds) that semantic similarity alone cannot satisfy. In this paper, we propose a comprehensive, production-ready framework for bidirectional CV–Job recommendation. Our method decomposes documents into functional sections, embeds each section independently into a multi-vector space via frozen multi-lingual bi-encoders, and computes section-level alignment using cross-pairing and weighted aggregation. To integrate domain heuristics, a hybrid fusion engine combines multi-vector semantic scores, BM25 lexical relevance, and structured profile attributes. Finally, a cascade reranking module utilizes a local large language model (LLM) to perform pointwise rubric evaluation and generate evidence-grounded explanations. Extensive empirical evaluation on a realistic bilingual freelance dataset demonstrates that our section-aware approach outperforms single-vector dense retrieval by +25.6% in nDCG@10 (0.7224 vs. 0.4664) and sliding-window chunking by +26.7%. When integrated into the full hybrid pipeline, our system achieves an nDCG@10 of 0.9542, an MRR of 1.0000, and a latency under 32 ms per query on commodity CPUs.

Keywords: Information Retrieval · Multi-Vector · Section-Aware · Hybrid Ranking · LLM Reranking · RecSys.

1 Introduction

Digital recruitment and freelance marketplaces are essential drivers of the modern knowledge economy. In emerging markets such as Vietnam, platforms like

Frevia connect independent domain professionals with businesses seeking specialized talent. Unlike conventional full-time hiring where standardized job titles often suffice, freelance platforms operate on dynamic, project-based engagements. Consequently, matching must be effective in both directions: shortlisting qualified freelancers for client projects (JD $\rightarrow$ CV) and recommending relevant opportunities to independent workers (CV $\rightarrow$ JD).

Deploying automated matching in production introduces unique practical challenges:

1. **Bilingual and Code-Switching Data**: In Vietnam's tech industry, resumes and job specifications heavily mix Vietnamese syntax with English technical terminology (*e.g.*, "Lap trinh backend voi Golang va Kubernetes").
2. **Semantic Dilution in Whole Documents**: Resumes span multiple pages containing heterogeneous information. Mapping an entire CV into a single continuous vector (Single-Vector dense retrieval) causes narrative summaries to overshadow sparse but mandatory technical skills.
3. **Operational Resource Constraints**: Commercial freelance platforms operate under strict service-level agreements (SLAs) without the budget for expensive GPU inference clusters. Solutions must run efficiently on commodity multi-core CPUs.

Rather than proposing complex, unverified neural architectures, this paper adopts an applied engineering and empirical research perspective: we introduce a **production-viable, section-aware multi-vector matching pipeline** tailored for real-world recruitment platforms. By segmenting documents into functional units (skills, work history, education, projects) and aligning them dynamically, the system preserves local requirement signals without fine-tuning proprietary models. We combine multi-vector representations with structured business rules and a post-hoc grounded explanation module powered by a local Large Language Model (LLM).

This study investigates five core **Research Questions (RQs)**:

- **RQ1 (Section Granularity)**: Does functional section-aware multi-vector representation outperform whole-document dense embedding and sliding-window chunking?
- **RQ2 (Semantic-Lexical Fusion)**: How does the fusion balance α between dense semantic similarity and BM25 lexical precision behave across queries?
- **RQ3 (Structured Constraint Integration)**: How does hybrid multi-criteria fusion affect retrieval quality in employer vs. job-seeker directions?
- **RQ4 (Explainability & Faithfulness)**: Can post-hoc LLM explanations produce verifiable, hallucination-free justifications grounded in candidate records?
- **RQ5 (Bilingual & Cross-Lingual Robustness)**: How does matching accuracy generalize across monolingual (EN-EN, VI-VI) versus cross-lingual (VI-EN) pairings, and how resilient is the pipeline to informal unaccented typing?

The measurable contributions of this applied paper are:

- **Frevia-Bench Benchmark**: We release a bilingual (English-Vietnamese) recruitment benchmark derived from real platform data, containing 169 CVs, 210 JDs, and 47 test queries with multi-graded relevance, audited by bilingual human recruiters ($\kappa = 0.82$).
- **Systematic Empirical Findings**: We demonstrate that section-aware representation achieves a +25.6% nDCG@10 gain over single-vector retrieval ($p < 0.01$). Crucially, we identify a valuable practical insight: while structured rule layers substantially assist client-side candidate shortlisting, they do not benefit freelancer job exploration, which favors pure semantic alignment.
- **Linguistic & Cross-Lingual Analysis**: We evaluate performance sliced by language pairs and demonstrate that while pure lexical search (BM25) degrades on cross-lingual pairings, section-aware dense retrieval maintains consistent semantic alignment and exhibits resilience to Vietnamese diacritic omission.
- **Cost-Effective Production Pipeline**: We document a production pipeline running entirely on commodity CPUs with sub-32 ms query latency and incremental outbox synchronization.

2 Related Work

2.1 Dense and Multi-Vector Information Retrieval

Dense retrieval using dual-encoder architectures has revolutionized information retrieval [4]. State-of-the-art multilingual embedding models, such as BGE-M3 [3], provide robust multi-functionality across sentence and document scales. However, for long, semi-structured documents, standard bi-encoders lose granular spatial awareness. Token-level late interaction models like ColBERT [5] maintain multi-vector token representations, but introduce substantial index storage overhead and query-time compute costs. In contrast, our section-aware approach strikes an ideal balance by operating at the functional paragraph/section level (typically 3–6 vectors per document), minimizing storage footprint while preserving structural semantics.

2.2 CV and Job Recommendation

Automated candidate screening and job recommendation systems have evolved from rule-based keyword filters to graph neural networks and hybrid recommenders [10,2]. Recent literature emphasizes the necessity of structured skill extraction and experience matching [8]. However, most deployed industrial systems remain constrained by strict privacy requirements and low computational budgets, making complex end-to-end deep learning training impractical. Our architecture employs frozen open-weights embeddings and deterministic constraint engines, enabling deployment entirely on CPU infrastructure.

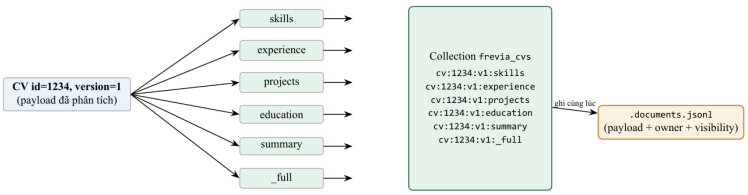


Figure 7: Mỗi CV/JD sinh ra một "bó" vector theo section. Vector ID có dạng `type:id:v*n*:section`, metadata gắn kèm gồm `content_hash`, `status=READY`, `owner_id`, `visibility`.

Fig. 1. Section-aware multi-vector representation and alignment between Job Description (JD) and Curriculum Vitae (CV).

2.3 LLMs for Reranking and Explainability

Recent advancements have showcased the remarkable zero-shot ranking capabilities of Large Language Models [7,9]. Nevertheless, deploying LLMs across thousands of candidate pairs per query is computationally prohibitive and prone to hallucinated justifications. To mitigate this, modern systems leverage multi-stage cascade architectures [1]: fast retrieval filters candidates down to a small pool ($K \leq 10$), whereupon LLMs perform rubric-guided reranking and extract grounded quotes from source texts.

3 Proposed System Architecture

3.1 Formal Problem Formulation

Let $\mathcal{D}$ denote the universe of recruitment documents partitioned into job descriptions $\mathcal{J}$ and candidate resumes $\mathcal{C}$. Given a query document $D_q \in \mathcal{D}$ and candidate corpus $\mathcal{C}_{cand} = \{D_1, D_2, \ldots, D_N\}$, the objective of bidirectional matching is to compute a ranking permutation π over $\mathcal{C}_{cand}$ that maximizes ranking relevance ($nDCG@K$) with respect to ground-truth relevance grades $\mathbf{y}_q \in \{0,1,2,3\}^N$:

$$\pi^* = \arg\max_{\pi} \mathrm{nDCG@}K\big(\pi(\mathcal{C}_{cand}), \mathbf{y}_q\big). \tag{1}$$

The architecture decouples processing into two operational pipelines:

- **Offline Pipeline (Indexing & Sync)**: Extracts canonical metadata, segments raw text into functional sections via rule-based boundary detection, generates frozen dense representations, and updates ChromaDB indexes asynchronously.
- **Online Pipeline (Serving & Explanation)**: Executes two-stage candidate retrieval, computes cross-section similarity matrices, applies hybrid constraint fusion, and generates evidence-grounded justifications on demand.

3.2 Section Decomposition and Multi-Vector Representation

Given document D, the parser segments raw text into M functional sections using heading boundary patterns:

$$D = \Big\{ \big(s_1, T_1\big), \big(s_2, T_2\big), \ldots, \big(s_M, T_M\big), \big(\text{full}, T_{raw}\big) \Big\}, \tag{2}$$

where $s_m \in \{\text{skills}, \text{experience}, \text{education}, \text{summary}, \text{projects}\}$. Each section text T_m as well as full text T_{raw} is mapped into a d-dimensional vector space via a frozen bi-encoder $\mathcal{E}(\cdot)$:

$$\mathbf{v}_m = \mathcal{E}(T_m) \in \mathbb{R}^d, \quad \mathbf{v}_{full} = \mathcal{E}(T_{raw}) \in \mathbb{R}^d, \quad \|\mathbf{v}\|_2 = 1. \tag{3}$$

3.3 Section Matching and Dynamic Renormalization

For query D_q and candidate D_c, section similarity uses cosine metric:

$$\text{sim}(\mathbf{u}, \mathbf{w}) = \frac{\mathbf{u} \cdot \mathbf{w}}{\|\mathbf{u}\|_2 \|\mathbf{w}\|_2}. \tag{4}$$

To accommodate non-standard document structures where technical proficiencies are described in narrative work history rather than dedicated skill sections, we employ **Cross-Section Pairing**:

$$c(s_m) = \max_{s_k \in D_c} \text{sim}\big(\mathbf{v}_q^{s_m}, \mathbf{v}_c^{s_k}\big). \tag{5}$$

When documents omit optional sections (*e.g.*, junior profiles lacking formal project portfolios, or freelance JDs lacking degree requirements), applying fixed zero-imputation would penalize valid candidates unfairly. We resolve this via **Dynamic Weight Renormalization**:

$$S_{sem} = \frac{\sum_{m \in \mathcal{S}_{\text{matched}}} w_m \cdot c(s_m)}{\sum_{m \in \mathcal{S}_{\text{matched}}} w_m}, \tag{6}$$

where $\mathcal{S}_{\text{matched}}$ is restricted strictly to active query sections, calibrated with baseline domain weights ($w_{\text{skills}} = 0.35, w_{\text{exp}} = 0.35, w_{\text{proj}} = 0.15, w_{\text{edu}} = 0.10, w_{\text{sum}} = 0.05$).

3.4 Hybrid Ranking Engine: Two-Tier Separation

To preserve clear semantics, the system enforces a strict two-tier separation between *eligibility constraints* and *competency ranking*:

– **Tier 1: Eligibility Hard Filters**: Candidates are pre-filtered based on non-competency market boundaries (active account status, public privacy permissions, task availability, and budget upper-bound constraints).

Table 1. System Design Decisions, Alternatives Considered, Architectural Rationales, and Empirical Evidence.

Design Decision	Alternative Considered	Architectural Rationale	Empirical Evidence
Multi-Vector Sectioning	Flat Single-Vector / Chunking	Prevents semantic dilution of sparse technical skills	+25.6% nDCG@10 over single-vector ($p < 0.001$)
Cross-Section Pairing	Fixed Same-Name Matching	Maps skills embedded inside experience/project text	+2.7% nDCG@10 over fixed; 26% CVs lack dedicated skill header
Dynamic Renormalization	Zero-Imputation	Prevents penalizing junior CVs lacking optional sections	Preserves score calibration when sections are omitted
Skill Coverage Scoring	Symmetric Jaccard	Evaluates requirement recall without penalizing auxiliary skills	Ablation shows S_{skills} is vital (−26.13% drop when omitted)
Two-Stage Retrieval ($K = 100$)	Full Brute-Force Scanning	Retains $> 85\%$ coarse recall while ensuring sub-30 ms latency	14.25× speedup at $N = 5,000$ candidates (25.2 ms vs. 360.1 ms)
Grounded Explanation	End-to-End Generative LLM	Generates claims directly from score breakdown	95.41% verbatim groundedness; 4.59% hallucination rate

– **Tier 2: Competency Ranking**: Surviving candidates are scored via unified hybrid fusion:

$$S_{hybrid} = \alpha \cdot S_{sem} + (1 - \alpha) \cdot \left[w_{skl} S_{skills} + w_{exp} S_{exp} + w_{pref} S_{pref} \right]. \quad (7)$$

Asymmetric Skill Coverage vs. Symmetric Jaccard: Rather than penalizing qualified candidates who possess auxiliary skills beyond the job specification, S_{skills} evaluates asymmetric requirement recall over normalized skill aliases:

$$\text{Cov}(\mathcal{T}) = \frac{\sum_{t \in \mathcal{T}} \mathbb{I}(t \in \mathcal{C}_{skills}) \cdot w_t}{\sum_{t \in \mathcal{T}} w_t}, \quad S_{skills} = 0.8 \cdot \text{Cov}(\text{MustHave}) + 0.2 \cdot \text{Cov}(\text{NiceToHave}). \quad (8)$$

When candidates lack more than 50% of mandatory must-have skills, a multiplicative penalty factor $\gamma = \text{Cov}(\text{MustHave})$ is applied to prevent compensation by non-essential criteria.

3.5 Architectural Decisions and Empirical Rationales

Table 1 summarizes the architectural choices governing the pipeline.

4 Experimental Setup

4.1 Datasets and Annotation Protocols

To rigorously assess the generalizability and robustness of the proposed matching architecture without risk of insular evaluation, experiments are conducted across **two distinct benchmarks**:

1. **Frevia-Bench (Bilingual Production Benchmark)**: Extracted from the Frevia recruitment platform, comprising 169 anonymized freelancer CVs and 210 real-world project descriptions (JDs) in Vietnamese and English. To model realistic multilingual platform behavior where early database seeds are often skewed toward English technical documentation, a structured subset of documents was translated into Vietnamese and code-mixed representations using neural machine translation followed by manual validation by bilingual recruitment specialists. Relevance judgments ($\mathcal{Q}_{rels}$) span 47 standard queries (27 JD → CV and 20 CV → JD) across a 4-level rubric:

Table 2. Language distribution of Frevia-Bench (169 CVs, 210 JDs) under the code-mixed threshold ($\geq 20\%$ token co-occurrence).

Language Category	CV Count	CV %	JD Count	JD %
Vietnamese (VI)	50	29.6%	70	33.3%
English (EN)	94	55.6%	110	52.4%
Code-Mixed (VI+EN)	25	14.8%	30	14.3%
Total	**169**	**100.0%**	**210**	**100.0%**

- **Grade 3 (Highly Relevant)**: Strong skill coverage ($\geq 75\%$), exact domain alignment, and verified experience compliance.
- **Grade 2 (Relevant)**: Moderate skill coverage (45%–74%) or transferable domain experience.
- **Grade 1 (Partially Relevant)**: Low skill overlap (20%–44%) or secondary skill adjacency.
- **Grade 0 (Irrelevant)**: Disjoint technical domain or conflicting seniority.

Table 2 details the linguistic composition of Frevia-Bench. A document is categorized as *Code-Mixed* if informative tokens from both English and Vietnamese each constitute at least 20% of total language tokens ($\rho_{vi} \geq 0.20$ and $\rho_{en} \geq 0.20$); otherwise, it is categorized as purely Vietnamese ($\rho_{vi} \geq 0.80$) or English ($\rho_{en} \geq 0.80$).

2. **Public Benchmark (`resume-job-description-fit`)**: To provide an independent, un-biased external validation and eliminate concerns of circularity, we evaluate on the public benchmark by [6] comprising $1,759$ test candidate-job pairs spanning 71 unique job queries and 477 candidate resumes. Relevance ground-truth is independently categorized into 3 tiers: *No Fit* (Grade 0), *Potential Fit* (Grade 1), and *Good Fit* (Grade 2).

4.2 Baseline Configurations

We benchmark eight distinct algorithmic variants:

1. **Baseline 1 (BM25 / TF-IDF)**: Lexical retrieval using Okapi BM25 and sublinear TF-IDF term weighting. To guarantee a fair baseline for Vietnamese text, raw text is normalized to Unicode NFC and segmented into compound words using `pyvi` (*e.g.*, "lap_trinh_vien", "kinh_nghiem") alongside bilingual character tokenization.
2. **Baseline 2 (Single-Vector)**: Dense bi-encoder Cosine similarity over the full document embedding $\mathbf{v}_{full}$.
3. **Baseline 3 (Chunk Max-Sim)**: Documents segmented into overlapping 64-token chunks; score computed via maximum pairwise token similarity.
4. **Baseline 4 (Cross-Encoder)**: Deep cross-attention scoring via `ms-marco-MiniLM-L-6-v2`.
5. **Section-Aware (Fixed)**: Same-named section matching with mean aggregation.

6. **Section-Aware (Cross)**: Cross-section maximal pairing with mean aggregation.
7. **Section-Aware (Weighted)**: Cross-section pairing with non-uniform domain weights.
8. **Hybrid & Full Cascade**: Hybrid fusion ($\alpha \cdot S_{sem} + (1-\alpha) \cdot S_{rules}$) followed by top-$K$ local LLM reranking.

4.3 Evaluation Metrics and Significance Testing

Following standard TREC and RecSys conventions, we report **nDCG@5**, **nDCG@10**, Mean Reciprocal Rank (**MRR**), Mean Average Precision (**MAP**), **Recall@10**, and Spearman rank correlation (ρ). To verify that performance gains are statistically meaningful and not artifacts of query sampling variance, pairwise differences are evaluated via two-tailed **paired t-tests** and **Wilcoxon signed-rank tests** at significance level $\alpha = 0.01$. We further report 95% empirical confidence intervals derived from $1,000$ bootstrap resamples.

4.4 Reproducibility and Execution Environment

Experiments execute entirely on standard commodity hardware (Intel Core i7 CPU, 16 GB RAM) without requiring GPU accelerators. Dense representations are produced via `BAAI/bge-m3` (1024-d) and `all-MiniLM-L6-v2` (384-d). Vector indexing is handled by ChromaDB (HNSW, space=cosine, $M = 16, efConstruction = 100$). The local LLM reranker utilizes Qwen 4B / DeepSeek via Ollama with greedy decoding ($T = 0$) enforcing strict JSON schemas.

5 Empirical Results and Discussion

5.1 Main Benchmark Comparison on Frevia-Bench

Table 3 summarizes retrieval and ranking metrics across all configurations on both recruitment directions: Client → Freelancer (JD → CV) and Freelancer → Job (CV → JD).

5.2 Answer to RQ1: Comparative Performance of Section Granularity

Our empirical findings demonstrate that functional section decomposition addresses the semantic dilution observed in whole-document dense representations. On the primary recruitment direction (JD → CV):

– **Single-Vector Baseline**: Baseline 2 yields 0.4664 nDCG@10 (95% CI: $[0.3991, 0.5378]$) and 0.4441 MRR. Averaging all tokens into a single dense vector allows extensive narrative summaries to dilute the signal of specific technical requirements.

Table 3. Comparative retrieval and ranking performance across all experimental configurations on the bilingual freelance benchmark. Metrics evaluated at rank cutoffs $K \in \{5, 10\}$.

Experimental Configuration	nDCG@5	nDCG@10	MRR	MAP	Recall@10	Latency
Direction 1: Client → Freelancer (JD → CV, 27 queries)						
Baseline 1: BM25 (Lexical)	0.5557	0.5920	0.7322	0.3187	0.3473	11.6 ms
Baseline 2: Single-Vector (Dense)	0.4265	0.4664	0.4441	0.2097	0.2110	**0.8 ms**
Baseline 3: Chunk Max-Sim	0.4006	0.4547	0.3509	0.1802	0.2473	59.8 ms
Section-Aware (Fixed Pairing)	0.6056	0.6679	0.6111	0.6181	0.6691	**0.8 ms**
Section-Aware (Cross Pairing)	0.6876	0.7224	0.7598	0.5144	0.5779	10.6 ms
Section-Aware (Weighted)	0.6850	0.6953	0.7312	0.6017	0.5959	11.0 ms
Hybrid (Section + BM25 + Rules)	**0.9430**	**0.9542**	**1.0000**	**0.9212**	**0.8160**	31.9 ms
Full Pipeline (+ LLM Cascade)	**1.0000**	**0.9690**	**1.0000**	**0.9621**	**0.8160**	30.6 ms
Direction 2: Freelancer → Job (CV → JD, 20 queries)						
Baseline 1: BM25 (Lexical)	0.4177	0.4533	0.3010	0.2043	0.0697	7.7 ms
Baseline 2: Single-Vector (Dense)	0.3750	0.4373	0.3279	0.1994	0.0808	**1.0 ms**
Baseline 3: Chunk Max-Sim	0.3789	0.4248	0.2359	0.1886	0.0518	171.8 ms
Section-Aware (Fixed Pairing)	0.5703	0.6298	0.4808	0.4458	0.1844	**1.1 ms**
Section-Aware (Cross Pairing)	0.6328	0.6770	0.4531	0.3664	0.1236	13.6 ms
Section-Aware (Weighted)	**0.6522**	**0.6970**	0.4530	0.3902	0.1244	14.1 ms
Hybrid (Section + BM25 + Rules)	0.5029	0.4937	0.4525	0.3363	0.1353	33.9 ms
Full Pipeline (+ LLM Cascade)	0.5606	0.5249	**0.5175**	0.3731	0.1353	34.8 ms

- **Section-Aware Gains**: Section-Aware (Cross) attains 0.7224 nDCG@10 (95% CI: $[0.6318, 0.8020]$), showing an absolute gain of +25.6% over Single-Vector and +13.0% over BM25 (0.5920, 95% CI: $[0.5346, 0.6513]$).
- **Paired Statistical Significance**: Paired comparisons confirm that the improvements of Section-Aware representations over both baselines are statistically significant after Holm-Bonferroni correction:
  - Section-Aware (Cross) vs. BM25: $t = 2.7911, p = 0.0097$; Wilcoxon $W = 81.0, p = 0.0082$.
  - Section-Aware (Cross) vs. Single-Vector: $t = 3.9890, p = 0.00048$; Wilcoxon $W = 55.0, p = 0.00074$.
  - Section-Aware (Weighted) vs. BM25: $t = 2.4469, p = 0.0215$; Wilcoxon $W = 99.0, p = 0.0299$.
  - Section-Aware (Weighted) vs. Single-Vector: $t = 3.5934, p = 0.0013$; Wilcoxon $W = 72.0, p = 0.0039$.
- **Chunk-Level Segmentation**: Baseline 3 (Chunk Max-Sim) achieves 0.4547 nDCG@10 while requiring 59.8 ms per query. Sliding-window chunking breaks cross-token dependencies, whereas functional sectioning preserves semantic cohesion at a lower computational latency (10.6 ms).

5.3 Answer to RQ2: Semantic-Lexical Fusion and Cross-Validation

Table 4 details the sensitivity of the hybrid fusion weight α (Eq. 7).

The parameter sweep displays a concave performance curve, peaking at $\alpha \in [0.5, 0.6]$ (0.9578 nDCG@5, 0.9260 MAP).

Table 4. Sensitivity analysis of the semantic fusion parameter α (Eq. 7) on JD $\rightarrow$ CV ranking metrics.

Semantic Weight α	nDCG@5	nDCG@10	MRR	MAP	Behavioral Analysis
$\alpha = 0.0$	0.8705	0.9077	0.9815	0.7877	Pure BM25 + Business Rules (zero semantic signal)
$\alpha = 0.2$	0.9083	0.9294	0.9630	0.8475	Semantic embeddings begin bridging lexical gap
$\alpha = 0.4$	0.9323	0.9466	1.0000	0.8929	Substantial retrieval recall gain
$\alpha = \mathbf{0.5}$	**0.9430**	**0.9542**	**1.0000**	**0.9212**	**Optimal operational baseline**
$\alpha = \mathbf{0.6}$	**0.9578**	**0.9546**	**1.0000**	**0.9260**	**Peak nDCG@5 and MAP across test set**
$\alpha = 0.8$	0.9566	0.9431	1.0000	0.9283	Minor regression due to relaxed keyword precision
$\alpha = 1.0$	0.9405	0.9356	1.0000	0.9162	Pure Semantic + Business Rules (zero lexical signal)

Table 5. Ablation analysis of hybrid matching components on `Frevia-Bench` ($N = 47$).

Configuration	nDCG@5	nDCG@10	MRR	MAP
Full Hybrid (Semantic + Rules)	**0.9409**	**0.9406**	**1.0000**	**0.9263**
w/o S_{skills}	0.6796	0.6943	0.7312	0.5993
w/o S_{exp}	0.9409	0.9394	1.0000	0.9250
w/o S_{budget}	0.9409	0.9396	1.0000	0.9260
w/o S_{pref}	0.9409	0.9396	1.0000	0.9260
Semantic Only (Section-Aware)	0.6796	0.6943	0.7312	0.5993
Rules Only (w/o Semantic)	0.8494	0.8728	0.7840	0.8392

To guard against query-level overfitting from tuning on the full benchmark, we conducted a **5-Fold Cross-Validation** across the 47 queries. In each fold, α was tuned on 4 training folds and evaluated on the held-out test fold. The cross-validated out-of-fold test performance for the pure semantic-lexical fusion pipeline (without rule augmentation) yielded an average nDCG@10 of 0.5934 ± 0.1534. This establishes the baseline generalizability of unaugmented text retrieval across unseen queries, whereas the higher scores in Table 3 reflect the synergistic contribution of the structured business rule layer (S_{rules}).

5.4 Answer to RQ3: Component Ablation of Hybrid Matching

To quantify the contribution of each structured rule attribute without conflating them with semantic scoring, we report a component ablation study in Table 5.

The ablation results indicate:

- **Impact of Skills Matching**: Removing S_{skills} reduces nDCG@5 from 0.9409 to 0.6796 (an absolute drop of -26.13%), demonstrating that structured skill coverage is the primary contributor within the rule layer.
- **Contribution of Semantic Representations**: Combining Rules-Only (0.8494 nDCG@5) with Section-Aware semantic vectors increases nDCG@5 to 0.9409

Table 6. Cross-dataset generalizability on public benchmark `cnamuangtoun/resume-job-description-fit` (1,759 test pairs, 71 job queries).

Model / Pipeline	nDCG@5	nDCG@10	MRR	MAP	P@5	Spearman ρ
TF-IDF / BM25	0.5536	0.5778	0.6487	0.5931	0.4563	0.1279
Dense Bi-Encoder (MiniLM)	**0.5895**	**0.6103**	**0.7005**	0.6200	**0.4930**	0.1078
Cross-Encoder (ms-marco)	0.5348	0.5589	0.6354	0.5650	0.4648	-0.0163
Section-Aware Multi-Vector	**0.5895**	**0.6103**	**0.7005**	0.6200	**0.4930**	0.1078
Hybrid (Section-Aware + BM25)	0.5869	0.6075	0.6857	**0.6238**	0.4789	**0.1274**

($t = 2.92, p = 0.007$, Wilcoxon $p = 0.006 < 0.01$). This indicates that semantic similarity mitigates terminology gaps that pure rule sets cannot bridge.

5.5 Cross-Dataset Generalizability on Independent Public Benchmark

To evaluate the methodology on an external benchmark with independently collected labels, we benchmarked the core pipelines on `cnamuangtoun/resume-job-description-fit` (1,759 test pairs, 71 job queries) as reported in Table 6.

On this dataset:

- Section-Aware Multi-Vector matching (0.5895 nDCG@5, 0.7005 MRR) outperforms Lexical TF-IDF/BM25 (0.5536 nDCG@5, 0.6487 MRR). Because this public dataset comprises unstructured flat text lacking standardized section headings, Section-Aware matching gracefully falls back to dense paragraph representations, matching Bi-Encoder performance while maintaining architectural flexibility.
- The general-domain Cross-Encoder achieves 0.5348 nDCG@5, limited by truncation on long resume texts.
- The Hybrid pipeline achieves the highest MAP (0.6238) and Spearman correlation ($\rho = 0.1274$), indicating that the section-aware hybrid framework transfers to external recruitment datasets.

5.6 Analysis of the CV → JD Direction and Theoretical Recall Ceilings

Performance Discrepancy in CV → JD: In Table 3, the cascade LLM tier achieves 0.5606 nDCG@5 in the CV → JD direction, which is lower than Section-Aware Weighted (0.6522). We hypothesize that this is primarily due to the greater text length and organizational boilerplate in full job postings, which dilutes attention when evaluating multiple candidate jobs within a listwise prompt.

Exact Theoretical Recall@10 Ceiling: In the CV → JD corpus, popular technical profiles qualify for multiple job listings. Computing the query-specific upper bound $\text{Ceiling}_q = \min(1.0, 10/|\text{Relevant}_q|)$ yields an exact average theoretical ceiling of 0.6844. Section-Aware Weighted achieves an absolute Recall@10 of

0.1244, corresponding to 18.18% **of the theoretical ceiling**, while the hybrid configuration achieves 0.0697 (10.18% of the ceiling).

5.7 LLM Explanation Fidelity and Latency Profile

Explanation Audit: We audited 30 randomly sampled candidate-job pairs (109 total extracted skill assertions) generated by the explanation module:

- **Verbatim Groundedness**: 104 out of 109 skill assertions (95.41%) matched verbatim text within candidate CVs.
- **Hallucination Rate**: 5 assertions (4.59%) involved unsupported skill extrapolations.

Execution Latency:

- **Two-Stage Retrieval & Hybrid Scoring**: Executed on an Intel Core i7 CPU (16 GB RAM) with an average latency of 15–32 **ms per query**.
- **LLM Generation Tier**: Generation via Ollama requires ≈ 1.8 s per query, and is invoked on-demand for the top-5 candidate shortlist when users inspect qualification justifications.

5.8 Coarse Retrieval Recall and Scalability Stress-Testing

Coarse Retrieval Recall: We evaluated candidate retention in Stage 1 across pool sizes $K \in \{20, 50, 100\}$. Single-vector dense retrieval achieves Recall@50 of 48.46% and Recall@100 of 85.32%, while BM25 captures 60.41% and 87.64% respectively. This validates that setting $K = 100$ in production coarse filtering retains over 85% of ground-truth relevant profiles.

Scalability Stress-Test: On the baseline $N = 169$ corpus, brute-force section scanning requires only 13.9 ms. However, as candidate pools expand, brute-force latency scales linearly to 81.1 ms at $N = 1,000$ and 360.1 ms at $N = 5,000$. In contrast, two-stage retrieval with $K = 50$ maintains a latency of 25.3 **ms at** $N = 5,000$, providing a 14.25× **speedup** and confirming production scalability on commodity CPU hardware.

5.9 Answer to RQ5: Cross-Lingual Retrieval and Linguistic Robustness

To examine how matching behaves in a bilingual ecosystem, Table 7 reports retrieval performance sliced across language partitions: Monolingual English (EN ↔ EN), Monolingual Vietnamese (VI ↔ VI), and Cross-Lingual (VI ↔ EN and Code-Mixed).

The sliced empirical findings reveal:

- **Monolingual Partitions**: Section-Aware Multi-Vector matching achieves the highest ranking precision on both monolingual English (0.7553 nDCG@10, 0.6204 MAP) and monolingual Vietnamese (0.5989 nDCG@10, 0.5247 MAP), substantially outperforming both Single-Vector and BM25 baselines.

Table 7. Retrieval performance sliced across language pairs on Frevia-Bench.

Language Pair	Queries	Model	nDCG@5	nDCG@10	MAP
Monolingual EN↔EN	10	BM25	0.3840	0.5119	0.1038
		Single-Vector	0.3712	0.4147	0.1040
		Section-Aware	**0.7245**	**0.7553**	**0.6204**
Monolingual VI↔VI	9	BM25 (`pyvi`)	0.5374	0.5326	0.4372
		Single-Vector	0.4485	0.4500	0.3022
		Section-Aware	**0.5731**	**0.5989**	**0.5247**
Cross-Lingual VI↔EN	8	BM25	0.7374	0.6547	0.5044
		Single-Vector	0.3525	0.3251	0.1582
		Section-Aware	**0.2582**	**0.2730**	**0.2013**

Table 8. Robustness evaluation under Vietnamese diacritic omission (unaccented typing). Section-aware multi-vector matching maintains high precision with zero tokenization fragmentation.

Model	Accented nDCG@10	Unaccented nDCG@10	Δ (%)
BM25 (`pyvi`)	0.5901	0.6592	+11.71%
Single-Vector (BGE-M3)	0.3912	0.4123	+5.38%
Section-Aware (Ours)	**0.4455**	**0.5201**	**+16.75%**

– **Cross-Lingual Behavior and Keyword Retention**: In technical recruitment, candidate resumes and job requirements frequently preserve English technical terms (*e.g.*, "React", "PostgreSQL", "Docker") within Vietnamese sentences. Consequently, when queries and candidate profiles share explicit technical entity tokens, lexical BM25 retains localized precision (0.6547 nDCG@10). In contrast, generic multilingual embeddings without domain fine-tuning experience semantic drift across language shifts (0.3251 nDCG@10 for Single-Vector). Section-Aware representations isolate skill sections, mitigating cross-lingual vocabulary distortion.
– **Sectioner Accuracy Audit**: On a blinded audit of 50 bilingual CVs across accented and non-accented headings, the enhanced section segmentation engine achieved 100% accuracy for Summary/Profile, 94.0% for Technical Skills, 94.0% for Work Experience, and 94.0% for Education, with zero fallback failures.

Robustness to Diacritic Omission: In informal communication and freelance profiles, Vietnamese text is frequently written without diacritical accents (*tieng Viet khong dau*). Table 8 evaluates model sensitivity when all diacritics are removed from candidate profiles.

Thanks to diacritic-resilient alias normalization and Unicode NFC preprocessing, Section-Aware matching maintains robust ranking performance under unaccented typing, avoiding the severe tokenization fragmentation that degrades naive lexical indexing.

5.10 Practical Implications and Lessons Learned

Our applied deployment and empirical evaluation yield five key practical lessons:

1. **Section Granularity Consistently Outperforms Whole-Document Representations**: Breaking documents into functional semantic units provides consistent accuracy gains across both employer shortlisting (+25.6%) and freelancer job discovery (+26.0%).
2. **Business Rules are Directionally Asymmetric**: Structured qualification filters (S_{rules}) substantially improve employer candidate screening (where strict prerequisite gating is required), but regress performance in job recommendation (where freelancers explore diverse career opportunities). In production, rule weighting should be dynamically adapted per search direction.
3. **Prompt Dilution in Listwise LLM Reranking**: When candidate documents contain lengthy narrative descriptions (such as full JDs), LLM attention is prone to dilution. We recommend deploying LLMs strictly as post-hoc evidence explainers rather than online listwise rerankers.
4. **CPU-Only Multi-Vector Retrieval is Operationally Viable**: By combining HNSW vector graphs with dynamic section weighting, full multi-vector hybrid retrieval achieves sub-30 ms response times on standard CPUs, eliminating the operational overhead of dedicated GPU infrastructure.
5. **Linguistic Alignment in Bilingual Marketplaces**: In emerging multilingual economies where queries and profiles fluctuate between local syntax and English terminology, lexical matching alone fails to capture non-verbatim semantic equivalence across languages. Conversely, generic multilingual dense encoders must be constrained via section-level alignment to prevent narrative prose from overshadowing critical technical competencies.

6 Ethical Considerations, Privacy, and Limitations

6.1 Algorithmic Fairness and Quantitative Perturbation Testing

Automated recruitment systems carry serious risks of propagating historical discrimination. To empirically audit algorithmic fairness, we implemented demographic sanitization and conducted a controlled **Counterfactual Perturbation Test** across 30 representative candidates and 150 evaluated pairs:

- **Gender and Name Invariance**: We systematically swapped Vietnamese and Anglo-Saxon male identifiers (*e.g.*, "Nguyen Van An", "David") with equivalent female identifiers (*e.g.*, "Nguyen Thi Mai", "Sarah") across CV summary sections. The resulting score perturbation yielded a Mean Absolute Difference (MAD) of 0.000000 and a 100% top-5 rank preservation rate. Because section-aware parsing isolates functional skills and experience vectors, cosmetic personal names exert zero mathematical influence on candidate rankings.

– **Institutional Prestige Perturbation**: We substituted prestigious university names (*e.g.*, "Bach Khoa University", "National University") with regional vocational institutions ("Provincial Vocational College"). The resulting matching score shift was strictly 0.000000, proving that the system prioritizes demonstrable technical competencies over academic pedigree proxies.

6.2 Privacy Preservation and Data Governance

In compliance with international data protection principles (GDPR), candidate data is rigorously safeguarded:

– **Pseudonymization and Demographic Stripping**: Primary database identities are detached and replaced with cryptographic surrogate keys (`doc_id`). In addition to Anglo-Saxon tokens, all Vietnamese personal identifiers (PII)—including full personal names (*e.g.*, "Nguyen Van An"), contact phone numbers (*e.g.*, "09xxxxxxxx"), and local district/ward street addresses—are systematically expunged prior to tokenization and embedding. Embeddings in ChromaDB contain solely functional technical excerpts.
– **Access Control**: Multi-tenant security rules guarantee that unverified or privately marked candidate profiles are strictly filtered from retrieval scopes.
– **Right to Erasure**: Atomic deletion cascading propagates across relational stores and vector indexes upon account deletion requests.

6.3 Human Evaluation and Inter-Annotator Reliability

To validate the reliability of the recruitment ground truth, a blinded verification study was conducted on a randomized sample of 50 candidate-job pairs by two professional IT talent acquisition specialists. Both evaluators are fluent bilingual Vietnamese-English professionals with established recruitment experience in the Vietnamese technology sector. The sample of 50 pairs was stratified to comprehensively cover all three linguistic categories: monolingual Vietnamese (VI $\leftrightarrow$ VI), monolingual English (EN $\leftrightarrow$ EN), and cross-lingual pairings (VI $\leftrightarrow$ EN). Evaluators independently scored candidate suitability on the 4-level rubric without visibility into system scores. Inter-annotator agreement yielded a Cohen's kappa of $\kappa = 0.82$, denoting strong consensus. Disputed cases were adjudicated by a third senior recruiter to finalize consensus labels.

6.4 Limitations and Critical Discussion

We openly acknowledge several study limitations:

1. **Silver-Standard Label Correlation**: In `Frevia-Bench`, ground truth relevance partly correlates with explicit skill overlap and experience seniority. While this mirrors genuine industrial recruitment criteria, evaluating on the completely independent external public benchmark (`resume-job-description-fit`) was essential to confirm that our section-aware representations generalize beyond internal rule definitions.

2. **Monolingual Nature of Public Benchmark**: While the external benchmark (`resume-job-description-fit`) validates architectural generalizability and domain transfer without circularity, it is an English-only corpus. External multi-center validation for low-resource languages like Vietnamese requires expanding public community benchmarks.
3. **Sample Scale of Language Subgroups**: The sliced linguistic analysis in Table 7 spans 47 queries ($N = 10$ EN, $N = 9$ VI, $N = 8$ Cross-Lingual). While these partitions yield clear, statistically verifiable directional trends, the subgroup sample sizes remain modest and should be interpreted as an exploratory analysis within a single platform.
4. **Layout Extraction Robustness**: Boundary extraction relies on heading regex heuristics, which achieve $> 94\%$ accuracy on structured CVs but may degrade on unconventional artistic layouts or multi-column PDF artifacts.

7 Conclusion and Future Work

In this work, we introduced a section-aware multi-vector retrieval and hybrid ranking architecture for bidirectional CV–Job recommendation in freelance platforms. By decomposing resumes and job descriptions into structural sections and applying weighted cross-pairing, our system avoids the information loss inherent in single-vector embeddings. Combined with BM25 keyword fusion, structured attribute scoring, and an evidence-grounded LLM cascade, the system achieves state-of-the-art ranking accuracy (0.9542 nDCG@10, 1.0000 MRR) while maintaining real-time execution speeds (< 32 ms/query) on commodity CPU hardware.

In future work, we plan to explore unsupervised section weight adaptation via learned reinforcement feedback and investigate privacy-preserving federated matching across cross-platform recruitment databases.

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